

# Zarin Tasnim

Software Engineer | Full-Stack Developer | AI/ML Developer

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## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, SQL, Java, Bash

**Backend:** Django, Django REST Framework, FastAPI, Flask, REST/JSON APIs, JWT Authentication, Stripe

**Frontend:** React, Vite, Tailwind CSS, HTML5, CSS3

**AI/ML:** PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI API, RAG, CNNs, Transfer Learning

**Databases:** PostgreSQL, MySQL, SQLite

**Tools:** Docker, Git/GitHub, Render, Vercel, Railway, Linux, pytest, Postman

## WORK EXPERIENCE

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**Full-Stack Developer (Freelance) | Baangi Bazaar** — Online Fashion Store | Remote **Feb 2026 – Present**

*Tech: Python, Django, DRF, React, PostgreSQL, JWT, Stripe, Docker, Render*

- Built and deployed a full-stack e-commerce platform supporting **20+ products** with size/color variants, inventory tracking, and a **4-step order management system** (Confirmed → Packed → Shipped → Delivered), enabling the business to reach **100–500 daily users**
- Engineered **15+ REST API endpoints** with **JWT authentication** and role-based access control; integrated **Stripe payment intents** for secure checkout and implemented product reviews/ratings with user-specific permissions
- Containerized the full application with **Docker**, deployed on **Render** with **PostgreSQL**, and built an admin dashboard for product upload, inventory control, order management, and sales overview

**Research Assistant | TSE Research Lab, Concordia University** | Montreal, QC **Sept 2021 – Jun 2025**

*Tech: Python, Java, Bash, Semantic Web, ontology modeling, ML*

- Designed an **ontology-driven semantic query model** aligned with modern **LLM** and **RAG**-based AI systems, reducing manual query formulation overhead across research workflows
- Co-authored **2 peer-reviewed IEEE publications** on latency reduction in holographic communication using In-Network Computing within **5G/6G** architectures
- Collaborated with researchers from **UQAM** and **Zayed University**; built **Python** and **Bash** automation scripts cutting experimentation overhead by **30%**

## PROJECTS

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**SeatLock — Movie Reservation System** | [Live](#) | [GitHub](#) **Aug – Sep 2025**

*Tech: Python, FastAPI, PostgreSQL, React, JWT, Docker, pytest, SQLAlchemy, Alembic, Render, Vercel*

- Built a full-stack movie reservation system supporting **8 movies** and **100+ seats** per showtime with **JWT authentication** and role-based access control (admin vs user)
- Implemented **PostgreSQL row-level locking** (SELECT FOR UPDATE) eliminating concurrent double-bookings across simultaneous users
- Containerized with **Docker**, achieved **15+ pytest tests**, and deployed on **Render + Vercel** with environment-based configuration

**URL Shortener with Analytics** | [GitHub](#) **Jul 2025**

*Tech: Python, FastAPI, PostgreSQL, SQLAlchemy, Chart.js, Railway*

- Built a **FastAPI** REST API with **4 endpoints** generating unique 6-character short codes with duplicate URL detection and input validation
- Implemented real-time click analytics dashboard using **Chart.js** showing top links sorted by popularity with pagination support
- Deployed on **Railway** with **PostgreSQL** backend; structured consistent JSON error handling across all endpoints

**Breast Cancer Detection** | [GitHub](#) **Jan – Apr 2022**

*Tech: Python, TensorFlow, PyTorch, VGG16, InceptionV3, ResNet50, scikit-learn, Matplotlib*

- Trained and compared **VGG16** (80%), **InceptionV3** (88%), and **ResNet50** (92%) using **transfer learning** on a balanced **4,000-image** subset from a **277K+** Kaggle histopathology dataset
- Applied **EarlyStopping** and dropout regularization with sigmoid output achieving **92% test accuracy** on binary IDC classification
- Evaluated all models with confusion matrices, **precision**, **recall**, and **F1-score**; identified ResNet50 as best performer with lowest false negative rate

## EDUCATION

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**M.Sc. Computer Science** | Concordia University, Montreal, QC

**Sept 2021 – Jun 2025**

**B.Sc. Computer Science** | BRAC University, Dhaka, Bangladesh

**Sept 2016 – Dec 2020**